

Notes: Dewatripont and Tirole (QJE, 1994)
**A Theory of Debt and Equity: Diversity of Securities and
Manager–Shareholder Congruence**

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1 Big picture

- **Research question.** Why do firms issue both debt and equity, why do different securities carry different control rights, and why are managers usually more aligned with shareholders than with debt-holders?
- **Main idea.** The financial structure of the firm is not only a way to divide cash flows. It is also a way to allocate control rights and to give outside claim-holders the right incentives to intervene in management.
- **Central friction.** Managers choose unobservable effort or projects, and later there may be noncontractible actions that outsiders can take. These actions can discipline managers but cannot be written directly into a complete contract.
- **Role of outsiders.** Since actions are noncontractible, the outsider who receives control must be given a payoff stream that makes him choose the desired action. Income rights and control rights therefore have to be designed together.
- **Main result.** Debt-like claims are naturally associated with control in bad states because their concave payoff makes their holders relatively tough. Equity-like claims are naturally associated with control in good states because their convex payoff makes their holders relatively soft.
- **Manager-shareholder congruence.** The model rationalizes why managers are partly aligned with equity-holders: both tend to prefer continuation and high-risk upside. Debt-holders, by contrast, are more willing to stop, liquidate, reorganize, or reduce risk.
- **Multiplicity of securities.** The model explains why a firm may need multiple outside investors and multiple securities. A controlling claim-holder must be incentivized to intervene, while another claim-holder may be needed to receive residual cash flows and balance the budget.
- **Economic takeaway.** Capital structure is a governance technology. Debt is a hard claim because it creates tough control in bad times, and equity is a soft claim because it leaves managers more freedom in good times.

2 Incremental contribution of the paper

- **Joint theory of income rights and control rights.** Earlier capital-structure models often studied cash-flow incentives or control rights separately. Dewatripont and Tirole show how the two must be matched.
- **Explains multiple securities.** The model rationalizes the coexistence of outside debt and outside equity rather than assuming that one outside claim-holder is enough.
- **Explains state-contingent control.** The paper predicts creditor control after bad performance and shareholder control after good performance. This matches the common intuition that creditors become important in distress, while equity is mostly passive in normal times.
- **Endogenizes manager-shareholder alignment.** Many earlier theories assume that managers act in shareholders' interests when making non-effort decisions. This paper provides a reason: the optimal incentive scheme tends to make managers' monetary incentives more equity-like.

- **Links security shape to intervention behavior.** A concave payoff, like debt, makes the controller more conservative and more willing to reduce risk. A convex payoff, like equity, makes the controller more tolerant of risky continuation.
- **Unifies corporate finance ideas.** The model gives foundations for themes in Jensen and Meckling, Ross, Myers and Majluf, and Jensen's free-cash-flow argument, but without simply assuming that the manager maximizes equity value.
- **Theoretical bridge.** The paper combines incomplete contracting, managerial moral hazard, security design, and corporate control into one formal framework.

3 Model and primitives

3.1 Managerial moral hazard

The model has two periods, $t = 1, 2$. In period 1, the manager chooses an unobservable effort level

$$e \in \{\underline{e}, \bar{e}\}, \quad \underline{e} < \bar{e}.$$

High effort is efficient but costs the manager K in utility terms. Low effort costs zero and can be interpreted as choosing a bad project or taking private benefits.

Firm profit in period t is denoted by π_t . Profits are verifiable. The first-period profit π_1 is informative about effort but is not a sufficient statistic. At the time π_1 is realized, investors also observe a signal u , whose distribution is informative about the future second-period profit π_2 .

Interpretation. Monetary pay alone cannot fully discipline the manager because some relevant information and actions are not contractible. Control rights therefore matter.

3.2 Interference in management

After effort is chosen and after π_1 and u are realized, the outside claim-holder who has control chooses an action

$$A \in \{S, C\}.$$

- C means acquiescing or continuing the manager's project.
- S means stopping, intervening, liquidating, reorganizing, reducing risk, or forcing a conservative course of action.

The action A is noncontractible. The capital structure therefore specifies who has the residual right to choose A for each realization of the first-period profit.

Interpretation. The paper's central governance problem is not merely who receives cash flows, but who gets to interfere with management after performance information arrives.

3.3 Risk structure

The model assumes that action S reduces the riskiness of second-period firm value relative to action C . Action C has fatter lower and upper tails, while action S is safer.

- If a decision maker has a convex payoff in π_2 , she values the upside tail and prefers C more often.
- If a decision maker has a concave payoff in π_2 , she dislikes downside risk and prefers S more often.
- A decision maker with a linear payoff chooses the ex post efficient action based on total firm value.

The signal u matters because action C becomes more attractive as u rises. There is an ex post efficient cutoff $\tilde{u}$ such that S is efficient for low signals and C is efficient for high signals.

Economic message. Once stopping is risk-reducing, payoff curvature maps naturally into governance behavior: debt is tough and equity is soft.

3.4 Managerial preferences

The manager dislikes outside interference. The paper gives two versions of this idea.

- **Private-benefit version.** If the project continues, the manager receives private benefit B . If the project is stopped, the manager receives no private benefit.
- **Monetary-incentive version.** The manager receives a compensation scheme based on firm performance. Because continuation produces a fatter upper tail, a manager with option-like or bonus-like pay tends to prefer continuation.

Interpretation. Whether managers dislike intervention because they lose private benefits or because they lose upside compensation, the model generates the same broad need: outsiders must sometimes be empowered to act against managers' wishes.

3.5 Timing

The model's timing is:

1. Contracts are written. The managerial incentive scheme and the capital structure are chosen.
2. The manager chooses effort.
3. First-period profit π_1 and signal u are realized.
4. A renegotiation stage may occur.
5. The outside party with control chooses action $A \in \{S, C\}$.
6. Second-period profit π_2 is realized and cash flows are distributed according to the contracts.

Interpretation. The key contracting problem is solved ex ante, but the important information about whether to intervene arrives only after contracts have been signed.

4 Methodology and main theoretical results

4.1 Optimal intervention rule

The paper first derives the intervention rule that would best discipline the manager. Let $x(\pi_1, u)$ be the probability that action C is chosen after first-period profit π_1 and signal u .

In the private-benefit benchmark, the problem is to minimize ex post inefficiency subject to giving the manager enough expected continuation rents to induce high effort. The solution has a cutoff form:

$$\begin{aligned} x(\pi_1, u) &= 0 & \text{if } u < u^*(\pi_1), \\ x(\pi_1, u) &= 1 & \text{if } u \geq u^*(\pi_1), \end{aligned}$$

where the cutoff $u^*(\pi_1)$ is decreasing in π_1 .

- Low u is bad news about effort, so it should trigger interference.
- Low π_1 is also bad news about effort, so for low π_1 the model requires interference for a larger range of signals.
- High π_1 is good news about effort, so the manager can be left alone for a larger range of signals.

Economic message. Poor current performance should make outsiders tougher. Good current performance should make outsiders more passive.

4.2 Private-benefit benchmark

In the private-benefit version, the manager receives B if action C is chosen and zero if action S is chosen. External interference works because it removes the manager's private benefit when performance information is bad.

The optimal rule is not simply the ex post efficient rule $u \geq \tilde{u}$. Instead, it distorts intervention to provide incentives:

- when π_1 is low, the threshold $u^*(\pi_1)$ is above the ex post efficient cutoff $\tilde{u}$, so there is excessive stopping;
- when π_1 is high, the threshold $u^*(\pi_1)$ is below $\tilde{u}$, so there is excessive continuation.

Interpretation. The incentive scheme deliberately makes the manager fear bad performance. This fear is valuable ex ante even though it can create ex post inefficiency.

4.3 Monetary incentives with renegotiation

The second version replaces private benefits with monetary incentives and allows perfect renegotiation. The manager's wage is based on verifiable profits, and the manager is rewarded for high second-period profits.

The optimal wage has an option-like structure. For each π_1 , there is a target $H^*(\pi_1)$ such that

$$w(\pi_1, \pi_2) = 1 \quad \text{if } \pi_2 \geq H^*(\pi_1),$$

$$w(\pi_1, \pi_2) = 0 \quad \text{otherwise.}$$

Both $u^*(\pi_1)$ and $H^*(\pi_1)$ decrease with π_1 .

- The manager is rewarded for high second-period profit.
- If first-period profit is low, the required second-period target is higher.
- Because action C has a fatter upper tail, a manager paid with high-profit bonuses prefers continuation to stopping.

Interpretation. The model endogenizes equity-like managerial compensation. The manager likes upside, so stopping becomes a punishment.

4.4 Outsider incentive schemes

A central step is to design the payoff of the outsider who gets control. The claim-holder in control must have an incentive to choose the desired action.

Suppose a controlling outsider owns a fraction of long-term debt and a fraction of equity. The payoff comparison between C and S depends on the relative debt and equity bias in that outsider's claim.

- A claim-holder with a balanced linear claim maximizes total firm value and chooses the ex post efficient cutoff $\tilde{u}$.
- A debt-biased claim-holder has a concave payoff and chooses S too often.
- An equity-biased claim-holder has a convex payoff and chooses C too often.

Economic message. The same payoff stream that gives an investor cash-flow rights also gives that investor a preference over corporate policy.

4.5 Implementation by debt and equity

The paper shows how the desired cutoff rule can be implemented using debt and equity.

Let D_1 be short-term debt. If first-period profit π_1 is below D_1 , control shifts to debt-holders. If π_1 is above D_1 , control remains with equity-holders. The threshold is chosen so that

$$D_1 = \bar{\pi}_1,$$

where $\bar{\pi}_1$ is the first-period profit level at which the desired incentive cutoff equals the ex post efficient cutoff.

- **Low first-period profit.** Debt-holders control. Their debt-like payoff makes them relatively tough, which is needed after bad performance.
- **High first-period profit.** Equity-holders control. Their equity-like payoff makes them relatively passive, which is appropriate after good performance.
- **Long-term debt.** The level of long-term debt D_2 adjusts with first-period profit to fine-tune the chosen cutoff.

Interpretation. The model rationalizes a familiar pattern: creditors become powerful in bad states, while shareholders retain control in good states.

4.6 Recapitalization implementation

The simple debt/equity control implementation can generate a discontinuity in long-term debt at the point where control switches from creditors to shareholders. The paper then describes a more realistic recapitalization mechanism.

In this alternative implementation:

- equity-holders may retain control after poor first-period performance by injecting money into the firm;
- recapitalization is attractive only for intermediate signals;
- when the signal is sufficiently bad, shareholders do not recapitalize and debt-holders intervene;
- when the signal is sufficiently good, shareholders are happy to let debt-holders choose continuation.

Economic message. The exact implementation can vary, but the robust prediction remains: bad performance increases the probability of tough debt-like intervention.

5 Interpretation and identification of the mechanisms

5.1 What the paper rationalizes

The theory rationalizes three stylized facts at the same time:

- firms issue multiple securities held by multiple outside investors;
- securities combine specific return streams with specific control rights;
- managers' preferences are more congruent with shareholders' preferences than with creditors' preferences.

Interpretation. These facts are not treated as separate anomalies. They arise together because managerial discipline requires outsider intervention, and outsider intervention requires incentive-compatible claims.

5.2 Why debt is hard

Debt-holders have a concave payoff. Once they are close to being repaid, they care more about protecting downside value than about increasing upside risk. Therefore, they prefer safer actions more often than a value-maximizing outsider would.

- This makes debt useful when the manager needs to be disciplined.
- It also makes debt costly because creditors can be too tough.

- The cost of debt in the model is excessive interference.

5.3 Why equity is soft

Equity-holders have a convex residual payoff. They receive the upside after debt is paid and therefore tolerate riskier continuation more often.

- This makes equity useful when the manager should be left alone after good performance.
- It also makes equity costly because shareholders can be too passive.
- The cost of equity in the model is excessive continuation or excessive passivity.

5.4 Why managers and shareholders are partly aligned

Managers often receive bonuses, stock options, or other rewards tied to upside firm performance. Since action C creates a fatter upper tail of profits, managers prefer continuation. Equity-holders also prefer continuation more often because their claim is convex.

Interpretation. The model does not assume that managers naturally maximize shareholder value. It explains why optimal incentive design can make them behave as if they are closer to shareholders than to debt-holders on many non-effort decisions.

5.5 Why multiple securities are needed

The controlling outsider's payoff is chosen to induce the desired action, not necessarily to maximize total firm value ex post. Once the controlling outsider and the manager are paid according to their incentive schemes, another outside investor may be needed to receive the residual cash flow.

Economic message. Multiple securities solve a moral-hazard-in-teams problem. They let the firm separately create incentives for management, incentives for the controller, and residual budget balance.

5.6 Relation to earlier capital-structure models

The paper revisits early capital-structure theories that assume managers act in shareholders' interests.

- **Jensen and Meckling.** Debt can discipline managers who otherwise choose policies favored by equity-like incentives.
- **Ross.** A positive relation between firm value and leverage can be reinterpreted as a relation between firm value and debt-like control.
- **Myers and Majluf.** The model gives a reason why managers' choices over financing may reflect shareholder-like incentives.
- **Jensen free cash flow.** Debt-like control can restrain managers from undertaking bad projects that increase upside payoffs or private benefits.

Interpretation. Earlier models may have been right to use manager-shareholder congruence as a reduced-form assumption, but this paper explains where that congruence can come from.

5.7 Limitations of the theory

- The model is highly stylized: two periods, binary effort, and binary action S versus C .
- The central debt-versus-equity prediction relies on the assumption that intervention reduces risk. If intervention increased risk, the assignment of hard and soft control could change.
- The implementation abstracts from many legal and institutional details of bankruptcy, covenants, renegotiation, and dispersed security ownership.
- The paper is theoretical, so it rationalizes patterns rather than estimating their magnitude.

6 Tables and figures

6.1 Table I: Economic circumstances and action S

Read:

- Table I gives concrete interpretations of the stopping or intervention action S .
- In financial distress or poor performance, S can mean liquidation, divestment, choosing a conservative option, reorganization, wage concessions, or cutting private benefits.
- In an R&D setting, S can mean stopping, reducing, or redirecting a project.
- In banking, S can mean reducing risk-taking or liquidating nonperforming loans.

Economic message: The abstract action S is meant to capture many real forms of creditor or outsider discipline. It is not only formal liquidation; it can also be a conservative change in policy.

6.2 Figure I: Timing of events

Read:

- Figure I lays out the sequence: contracting, managerial effort, realization of π_1 and u , renegotiation, outsider action, and second-period profit.
- The manager chooses effort before the signal and first-period profit are realized.
- Control matters because action A is chosen only after information arrives and cannot be completely specified in the original contract.

Economic message: The timing shows why capital structure can discipline management: it determines who acts after interim performance reveals information about managerial behavior.

TABLE I

Economic circumstance	Action S
Financial distress (illiquidity) or poor performance	Liquidate, divest, choose conservative option, go on a diet (reorganization, wage concessions, reduction of private benefit)
Lack of innovation (R&D firm)	Stop project, reduce its size, or redirect it
Nonperforming loans (bank)	Reduce risk taking, liquidate

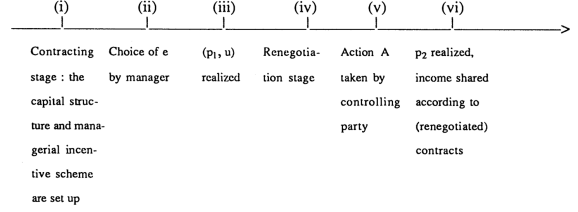
Table I: Economic circumstances and examples of the intervention action S .FIGURE I
Timing of Events

Figure I: Timing of contracting, effort, information, control, and payoff realization.

6.3 Figure II: Optimal interference rule

Read:

- Figure II plots the signal u against first-period profit π_1 .
- The downward-sloping boundary $u^*(\pi_1)$ separates stopping from continuation.
- Below the boundary, $x_1(\pi_1, u) = 0$, so action S is chosen.
- Above the boundary, $x_1(\pi_1, u) = 1$, so action C is chosen.
- Since the boundary slopes downward, low first-period profit requires a higher signal before the manager is allowed to continue.

Economic message: The optimal incentive scheme is harsher after bad performance and softer after good performance.

6.4 Figure III: Debt control versus equity control

Read:

- Figure III overlays the incentive-efficient partition with the ex post efficient partition.
- The horizontal line at $\tilde{u}$ is the ex post efficient cutoff.
- The downward-sloping $u^*(\pi_1)$ is the incentive-efficient cutoff.
- For low π_1 , the incentive-efficient cutoff is above $\tilde{u}$, so the firm uses debt control and accepts excessive toughness.
- For high π_1 , the incentive-efficient cutoff is below $\tilde{u}$, so the firm uses equity control and accepts excessive passivity.

Economic message: Debt and equity are assigned to different states because they create different biases in intervention.

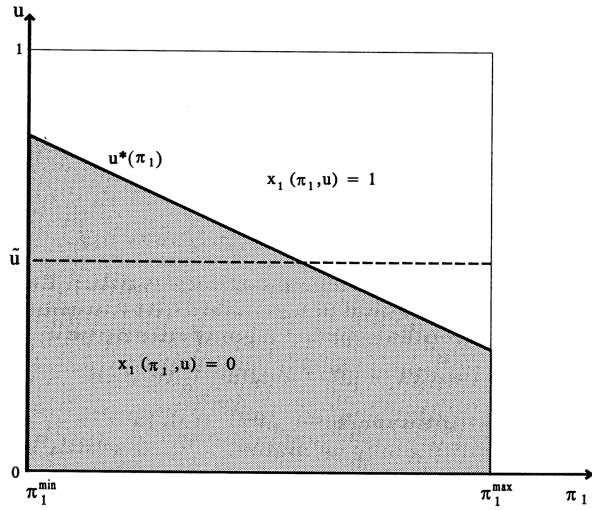


FIGURE II

Figure II: Optimal stopping/continuation rule as a function of π_1 and u .

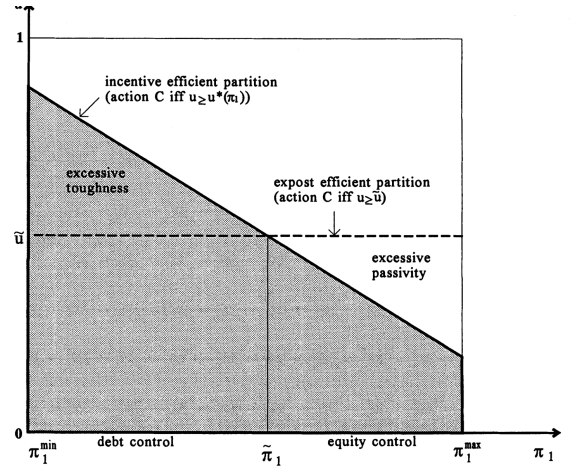


FIGURE III

Figure III: Debt control in bad states and equity control in good states.

6.5 Figure IV: Long-term debt and current performance

Read:

- Figure IV shows how the level of long-term debt D_2 varies with first-period profit π_1 in the pure debt/equity implementation.
- Within each control region, better first-period performance allows more long-term debt.
- At the control-switch point, there is a discontinuity in D_2 .
- This discontinuity reflects the abrupt transfer of control from debt-holders to equity-holders.

Economic message: Debt levels are not only financing quantities. They are used to tune the controlling investor's incentives.

6.6 Figure V: Recapitalization and smoother implementation

Read:

- Figure V shows a mechanism in which debt-holders have a conservative decision rule at a given leverage level.
- Equity-holders can recapitalize the firm in some intermediate states to avoid debt-holder stopping and preserve continuation.
- This recapitalization mechanism implements the same desired intervention rule without relying on a discontinuous jump in long-term debt.

Economic message: The exact institutional mechanism can vary, but the core logic is robust: poor performance raises the threat of creditor control, and shareholders may have to inject funds

to avoid that control.

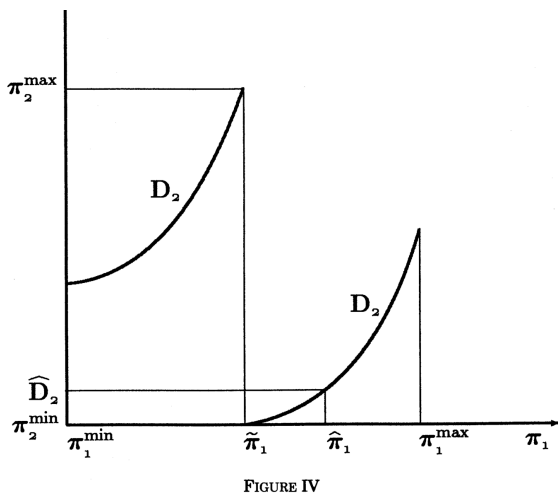


Figure IV: Long-term debt as a function of first-period profit.

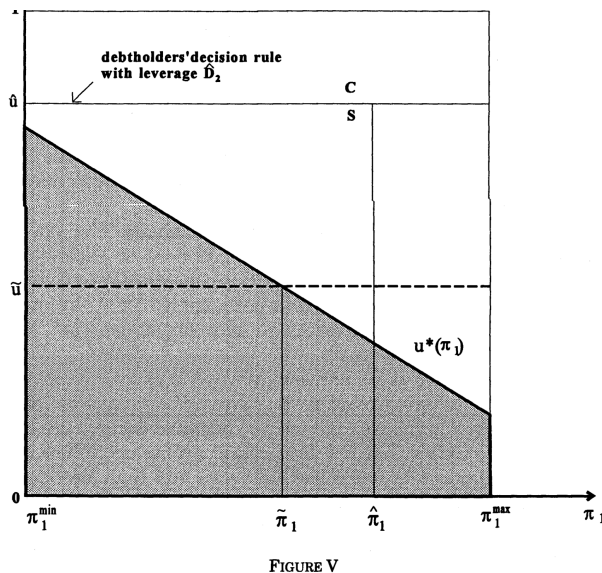


Figure V: Recapitalization mechanism that preserves the desired control rule.

7 Short summary

- The paper develops a theory of debt and equity based on managerial moral hazard and incomplete contracts.
- Managers dislike outside interference, either because it removes private benefits or because it reduces their upside compensation.
- Outsider control is useful because it can discipline managers after performance information arrives.
- Since the relevant actions are noncontractible, the controlling outsider must be given a payoff stream that induces the desired action.
- The optimal rule makes interference more likely after poor first-period performance and less likely after good first-period performance.
- Debt-like claims are concave and make claim-holders tougher; equity-like claims are convex and make claim-holders softer.
- The model predicts debt-holder control in bad states and equity-holder control in good states.
- This explains why debt is a hard claim, why equity is a soft claim, and why managers are more congruent with shareholders than with creditors.
- Multiple securities are needed because incentive provision for the manager and the controlling outsider generally requires a residual claimant to balance the accounts.
- The paper gives foundations for earlier capital-structure models that assume managers act in shareholders' interests.

8 Conclusion

8.1 Core conclusion

Dewatripont and Tirole show that capital structure is a mechanism for governance and incentives. Debt and equity are useful not merely because they divide cash flows differently, but because they assign different control rights to investors with different incentives to intervene.

8.2 Economic intuition

The manager needs to be disciplined after bad performance but should not be constantly interfered with after good performance. Debt-holders, because of their concave payoff, are naturally tough monitors in bad states. Equity-holders, because of their convex payoff, are naturally passive monitors in good states. The optimal capital structure uses both biases.

8.3 Incremental contribution revisited

The paper explains three facts together: the diversity of securities, the link between payoff shape and control rights, and the partial congruence between managers and shareholders. The key insight is that the security held by the controlling outsider is also that outsider's incentive scheme.

8.4 Implications

The model implies that the choice between debt and equity should be understood as a choice over who controls the firm in different states. More debt increases the probability of tough creditor intervention. More equity increases managerial freedom but can create excessive passivity.

8.5 Limitations

The model is deliberately stylized and does not estimate empirical magnitudes. It also relies on the assumption that intervention reduces risk. Real-world bankruptcy, covenants, renegotiation, and dispersed ownership introduce additional complications that are outside the model.

8.6 Takeaway

The enduring takeaway is that securities are bundles of cash-flow rights and control rights. Debt disciplines managers by threatening tough intervention in bad times; equity supports managerial initiative by providing softer control in good times.